Anaerobic Meningitis (MONDO:0002000): A Comprehensive Disease Characterization

Category: Infectious Disease | **Disease type:** Acquired bacterial CNS infection (non-heritable)
Evidence base: literature synthesis (no primary datasets provided). PMIDs cited throughout; evidence type noted as human clinical, animal/model, or in vitro.

Summary

Anaerobic meningitis is a rare, acquired infection of the leptomeninges caused by obligate or aerotolerant anaerobic bacteria, accounting for only about 2.4% of adult culture-proven bacterial meningitis. It is not a genetic or heritable disorder; instead it is a mechanistically driven infectious disease that arises almost exclusively in hosts with an anatomic or immunologic predisposition. The dominant routes of meningeal invasion are (i) contiguous spread from ear/nose/throat (ENT), odontogenic, or sinus foci; (ii) direct inoculation via neurosurgery, penetrating trauma, or indwelling CNS hardware (ventriculoperitoneal shunts, external ventricular drains); and (iii) hematogenous seeding or gut translocation. The principal causative organisms are Gram-negative anaerobes *Fusobacterium* (*necrophorum*, *nucleatum*), *Bacteroides fragilis*, and *Prevotella* species, together with Gram-positive anaerobes *Cutibacterium (Propionibacterium) acnes* and *Parvimonas micra (Peptostreptococcus micros)*, with mixed/polymicrobial infection in nearly half of adult cases.

The pathophysiology follows the canonical bacterial-meningitis inflammatory cascade: after anaerobes reach the subarachnoid space, bacterial components are recognized by Toll-like receptors, activating NF-κB and MAPK signaling, driving a surge of proinflammatory cytokines (TNF-α, IL-1β, IL-6) and chemokines, recruiting neutrophils that generate superoxide and nitric oxide, producing peroxynitrite-mediated oxidative stress, blood-brain-barrier breakdown, and ultimately neuronal injury. Pathogen-specific virulence factors amplify this process — notably the *B. fragilis* capsular polysaccharide (a direct abscess-forming, antiphagocytic factor) and the *F. necrophorum* leukotoxin.

Because fastidious anaerobes are frequently missed by routine culture, diagnosis increasingly depends on 16S rRNA PCR and metagenomic next-generation sequencing (mNGS), supported by CSF lactate (highly discriminatory for bacterial vs. aseptic meningitis) and neuroimaging for complications (mastoiditis, abscess, cavernous sinus/internal jugular thrombosis). Treatment combines metronidazole (excellent CSF penetration, bactericidal against obligate anaerobes) with a broad-spectrum β-lactam, plus surgical source control and adjunctive corticosteroids. Despite therapy, adult mortality is ~30%, and the majority of survivors have neurological sequelae, making this a high-morbidity condition even though it is uncommon.

Disease Information (Section 1)

Overview. Anaerobic meningitis is inflammation of the meninges caused by anaerobic bacteria. It is best regarded as a rare microbiological subset of bacterial meningitis rather than a distinct nosological entity. It is defined by the recovery (by culture or molecular methods) of obligate/aerotolerant anaerobes from cerebrospinal fluid (CSF) in a patient with clinical and CSF evidence of meningitis.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0002000 (anaerobic meningitis)
MeSH	"Meningitis, Bacterial" (D016920); anaerobic infection is not separately termed
ICD-10	G00–G03; specifically G00.8 (other bacterial meningitis), G03.8/G03.9 for unspecified/other
ICD-11	1D01 (bacterial meningitis)
SNOMED CT	Bacterial meningitis (concept)
OMIM / Orphanet	Not applicable — no Mendelian/rare-genetic entry (acquired infection)

Synonyms / alternative names. Anaerobic bacterial meningitis; meningitis due to anaerobic bacteria; organism-specific descriptors such as *Fusobacterium* meningitis, *Bacteroides fragilis* meningitis, *Cutibacterium/Propionibacterium acnes* meningitis, *Prevotella* meningitis, and *Parvimonas/Peptostreptococcus* meningitis.

Information source. The evidence base derives from a mixture of individual case reports/case series (EHR-derived clinical data) and aggregated single-center/national cohort studies; there are no disease-level genetic resources because the condition is not heritable.

Etiology (Section 2)

Primary cause — infectious/mechanistic. Anaerobic meningitis is caused by anaerobic bacteria that are normal commensals of the oral cavity, gut, and skin gaining access to the subarachnoid space. In a 30-year single-center series of 540 culture-proven adult bacterial meningitis cases, 13 (2.4%) were anaerobic, from which 14 strains were isolated: *Fusobacterium nucleatum* (3), *Prevotella* spp. (3), *Bacteroides fragilis* (3), *Cutibacterium/Propionibacterium acnes* (3), and *Parvimonas micra/Peptostreptococcus micros* (2) (PMID: 29396061).

> "including nine Gram-negative (G(-)) strains: Fusobacterium nucleatum (3), Prevotella species (3) and Bacteroides fragilis (3), and five Gram-positive (G(+)) strains: Propionibacterium acnes (3) and Peptostreptococcus micros (also known as Parvimonas micra) (2)" — PMID: 29396061

Risk factors (environmental / acquired). There are essentially no *genetic* causal risk factors. The dominant risk factors are anatomic and clinical:

- **Postneurosurgical state** — 53.8% (7/13) in the adult series (PMID: 29396061).
- **Underlying medical conditions** — 79.6% (10/13): diabetes mellitus, malignancy, liver cirrhosis, cerebral infarct, alcoholism (PMID: 29396061).
- **Head trauma, prior neurosurgical procedures, and implanted devices** (VP shunts, ventricular drains) (PMID: 32717475).
- **ENT infections** — near-universal precursor in pediatric *F. necrophorum* meningitis (acute otitis media in 21/27) (PMID: 41479136).
- **Odontogenic sources** — recent molar extraction, periodontitis (PMID: 41330476, PMID: 35692070).

> "53.8% (7/13) had a postneurosurgical state as the preceding event, and 79.6% (10/13) had underlying medical conditions including diabetes mellitus, malignancy, liver cirrhosis, cerebral infarct and alcoholism" — PMID: 29396061

Genetic modifiers (not causal). Host genetic variation modifies the *inflammatory response and outcome*, not susceptibility to infection per se: single-nucleotide polymorphisms in inflammasome genes **CARD8 (rs2043211)** and **NLRP1 (rs11621270)** are associated with poor outcome in bacterial meningitis (PMID: 23053059).

Protective factors. Prompt treatment/source control of predisposing ENT, dental, and abdominal infections; strict neurosurgical/device asepsis. No genetic protective alleles are established. Anticapsular antibody protects against *B. fragilis* bacteremia while T-cell-mediated immunity protects against abscess in animal models (PMID: 549184).

Gene-environment interactions. The only demonstrable GxE interplay is that inflammasome/complement genotype (CARD8, NLRP1; MASP-2) shapes how severely the host responds to the anaerobic infectious trigger (PMID: 23053059, PMID: 28086930).

Phenotypes (Section 3)

Clinical phenotypes are those of bacterial meningitis generally, though anaerobic cases are more likely to be subacute or atypical. In 696 adults with community-acquired bacterial meningitis, the classic triad of fever, neck stiffness, and altered mental status was present in only 44%, but 95% had ≥2 of the 4 cardinal features; 14% were comatose and 33% had focal neurologic deficits (PMID: 15509818).

> "The classic triad of fever, neck stiffness, and a change in mental status was present in only 44 percent of episodes; however, 95 percent had at least two of the four symptoms of headache, fever, neck stiffness, and altered mental status." — PMID: 15509818

Phenotype	Type	HPO suggestion	Frequency / notes
Fever	Symptom	HP:0001945	Common; may be absent in host-defective/atypical cases
Headache	Symptom	HP:0002315	Very common; may be subacute in anaerobic cases
Neck stiffness / meningismus	Clinical sign	HP:0031179 (nuchal rigidity)	Part of triad; absent in ~half
Altered mental status	Symptom/ sign	HP:0011446	Part of triad; 14% comatose
Seizures	Clinical sign	HP:0001250	Reported, esp. with abscess
Focal neurologic deficit	Sign	HP:0034332	~33% in bacterial meningitis
Cranial nerve palsy (e.g., VI)	Sign	HP:0006802 / HP:0000651	With cavernous sinus thrombosis (PMID: 41330476)
Photophobia	Symptom	HP:0000613	Reported (PMID: 41330476)
CSF pleocytosis (neutrophilic)	Lab abnormality	HP:0025432 (CSF pleocytosis)	Characteristic; occasionally absent in immunocompromised
Elevated CSF protein / low glucose	Lab abnormality	HP:0002922 / HP:0032672	Typical bacterial pattern
Elevated CSF lactate	Lab abnormality	—	Discriminatory marker

Onset/severity/progression. Age of onset is bimodal in practice — a pediatric cluster (median 4 years, *F. necrophorum*, ENT-driven) and adults (22–77 years), plus rare neonatal *B. fragilis* disease. Severity is typically severe; progression is acute to subacute. Anaerobic-specific atypical presentations include subacute headache without fever/neck stiffness in otogenic polymicrobial cases (PMID: 37629718) and cranial-nerve palsy with cavernous sinus thrombosis (PMID: 41330476).

Quality of life impact. Substantial — the majority of survivors carry neurological deficits (see Prognosis).

Genetic / Molecular Information (Section 4)

Not applicable as a heritable disease. Anaerobic meningitis has **no causal genes, no pathogenic germline variants, no inheritance pattern, no penetrance, and no carrier frequency** (PMID: 29396061; acquired-infection framing). The only host-genetic contribution is as an *outcome modifier*:

- **CARD8** (rs2043211) and **NLRP1** (rs11621270) inflammasome polymorphisms — associated with poor outcome (PMID: 23053059).
- **MASP-2** (complement lectin pathway) — higher CSF levels correlate with poor functional outcome (PMID: 28086930).

> "single-nucleotide polymorphisms in the inflammasome genes CARD8 (rs2043211) and NLRP1 (rs11621270) are associated with poor disease outcome" — PMID: 23053059

The relevant "molecular" genetics are microbial: bacterial virulence loci encoding the *B. fragilis* capsular polysaccharide biosynthesis genes and the *F. necrophorum* leukotoxin operon (see Mechanism).

Environmental Information (Section 5)

Environmental / lifestyle contributors operate by predisposing to anaerobic invasion: chronic otitis media, sinusitis, periodontal disease, dental extraction, alcoholism, diabetes mellitus, malignancy, cirrhosis, and neurosurgery/indwelling devices (PMID: 29396061, PMID: 32717475, PMID: 41330476).

Infectious agents (NCBI Taxonomy).

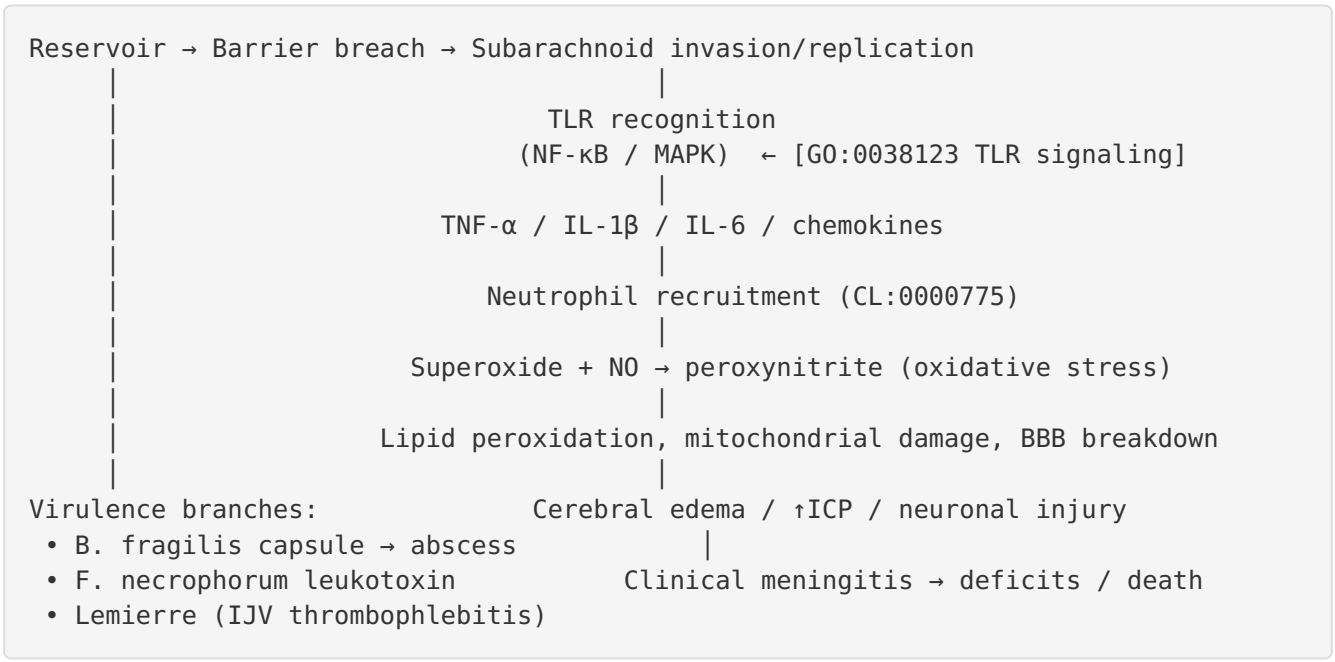
Organism	NCBI Taxon	Gram	Typical source
Fusobacterium necrophorum	txid859	–	Oropharynx/ENT; Lemierre
Fusobacterium nucleatum	—	–	Oral; odontogenic
Bacteroides fragilis	txid817	–	Gut; neonatal/abdominal
Prevotella spp.	—	–	Oral/ENT
Parvimonas micra	—	+	Oral
Cutibacterium acnes	txid1747	+	Skin; neurosurgical device
Clostridium perfringens	txid1502	+	Trauma/gut

Mechanism / Pathophysiology (Section 6)

Ordered causal chain

1. Endogenous anaerobes colonize a mucosal/skin site (oropharynx, gut, skin) → serve as the reservoir.
2. A predisposing lesion (ENT/odontogenic infection, neurosurgery, trauma, device, immunosuppression) **breaches** the normal barrier → **leads to** anaerobic access to blood or contiguous CNS tissue.
3. Anaerobes reach the subarachnoid space by contiguous spread, direct inoculation, or hematogenous/gut translocation → **results in** subarachnoid bacterial replication (favored by the immune-privileged, complement-poor CSF).
4. *Branch (F. necrophorum):* oropharyngeal infection → septic internal jugular vein thrombophlebitis (**Lemierre syndrome**) → septic emboli → CNS seeding (PMID: 42568628, PMID: 41479136).
5. *Branch (B. fragilis):* capsular polysaccharide → abscess formation independent of other organisms (PMID: 6954635).
6. Bacterial components are recognized by **Toll-like receptors** on antigen-presenting/brain cells → **activates** NF-κB and MAPK pathways (PMID: 22618789).
7. Transcriptional activation → **up-regulates** proinflammatory mediators **TNF-α, IL-1β, IL-6**, and chemokines (PMID: 22618789, PMID: 24105840).

8. Cytokine/chemokine surge → **recruits and activates** polymorphonuclear neutrophils into the subarachnoid space.
9. Activated neutrophils → **release** superoxide anion and nitric oxide → **form** peroxynitrite → **generate** oxidative stress (PMID: 22618789).
10. Oxidative stress → **causes** lipid peroxidation, mitochondrial damage, and **blood-brain-barrier breakdown** (PMID: 22618789).
11. BBB breakdown + inflammation → **produces** cerebral edema, raised intracranial pressure, impaired cerebral blood flow, and **neuronal injury** → clinical meningitis, deficits, death.
12. Host genetic modifiers (CARD8/NLRP1 inflammasome, MASP-2 complement) **tune** the magnitude of steps 5–9 and thereby outcome (PMID: 23053059, PMID: 28086930).



Supporting detail

> "S. pneumoniae is recognized by antigen-presenting cells through the binding of Toll-like receptors inducing the activation of factor nuclear kappa B or mitogen-activated protein kinase pathways" — PMID: 22618789
>
> "polymorphonuclear are attracted, activated and released in large amounts of superoxide anion and nitric oxide, leading to the peroxynitrite formation, generating oxidative stress. This cascade leads to lipid peroxidation, mitochondrial damage, blood-brain barrier breakdown" — PMID: 22618789

Virulence factors. The *B. fragilis* **capsular polysaccharide (CP)** is the key virulence factor: encapsulated strains form abscesses directly attributable to CP, and CP protects against phagocytosis (PMID: 6954635, PMID: 3865350). By contrast, *B. fragilis* **LPS is chemically incomplete and weakly endotoxic** — the abscess-inducing LPS dose is ~5× higher than that of CP — explaining infrequent DIC/purpura in *Bacteroides* sepsis (PMID: 549184, PMID: 3711690).

> "The abscesses caused by encapsulated strains were shown to be directly attributable to the capsular polysaccharide, which is an important virulence factor of this organism." — PMID: 6954635
>
> "the lipopolysaccharides of Bacteroides lack the biologic potency characteristic of endotoxin" — PMID: 549184

Model-organism corroboration. In Wistar rats, experimental Gram-negative (Klebsiella) meningitis produced a TNF-α/IL-1β/IL-6/CINC-1 surge within 6–24 h and long-term memory impairment (PMID: 24105840).

Ontology suggestions. GO:0038123 (Toll-like receptor signaling), GO:0006954 (inflammatory response), GO:0006979 (response to oxidative stress), GO:0097529 (myeloid leukocyte migration); CL:0000775 (neutrophil), CL:0000129 (microglial cell), CL:0000540 (neuron); CHEBI:16240 (hydrogen peroxide), CHEBI:18421 (superoxide), CHEBI:25812 (peroxynitrite); metronidazole (CHEBI:6909).

Anatomical Structures Affected (Section 7)

- **Primary organ / body system:** the **leptomeninges** (pia-arachnoid) and subarachnoid space of the **central nervous system** — UBERON:0002360 (meninx), UBERON:0002361 (pia mater), UBERON:0002362 (arachnoid mater), UBERON:0001359 (CSF), UBERON:0001017 (CNS).
- **Secondary involvement:** brain parenchyma (cerebritis/abscess, UBERON:0000955), ventricles (ventriculitis), cranial nerves, cavernous sinus and internal jugular vein (thrombosis), mastoid air cells and paranasal sinuses (contiguous foci).
- **Tissue/cell level:** vascular endothelium of the BBB, neutrophils (CL:0000775), microglia (CL:0000129), astrocytes (CL:0000127), neurons (CL:0000540).

- **Subcellular:** mitochondria (GO:0005739; oxidative damage), plasma membrane lipids (peroxidation), nucleus (NF-κB translocation).
- **Localization / lateralization:** the meningitis itself is typically diffuse/bilateral; complications (mastoiditis, subperiosteal/epidural abscess, IJV thrombosis, cavernous sinus thrombosis) may be unilateral or bilateral (PMID: 41479136, PMID: 42568628).

Temporal Development (Section 8)

- **Onset:** acute to subacute; anaerobic cases skew more subacute/insidious than pyogenic aerobic meningitis, sometimes without fever or neck stiffness (PMID: 37629718).
- **Age of onset:** pediatric cluster (median 4 y for *F. necrophorum*), adults 22–77 y, and rare neonates (*B. fragilis*) (PMID: 41479136, PMID: 29396061, PMID: 31838318).
- **Progression:** rapid if untreated; complicated meningitis (20/27 pediatric cases) with mastoiditis, subperiosteal/epidural abscess, or Lemierre syndrome, 12/27 requiring surgical drainage (PMID: 41479136).
- **Course/duration:** typically requires prolonged therapy (often 4–8 weeks when abscess present); not relapsing-remitting, though superinfection/recurrence occurs with retained devices (PMID: 28089421).
- **Critical period for intervention:** early antibiotic + corticosteroid administration (steroids before/with antibiotics) is the key therapeutic window (PMID: 41675892, PMID: 40113367).

Inheritance and Population (Section 9)

Inheritance: none — acquired infectious disease; inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effects, consanguinity, and carrier frequency are all **not applicable**.

Epidemiology.

Metric	Value	Source
Fraction of adult culture-proven bacterial meningitis	~2.4% (13/540 over 30 y)	PMID: 29396061
Positive anaerobic CSF cultures (5-y study)	24/594 (4%), only 3 clinically true	PMID: 32717475
Pediatric *F. necrophorum* meningitis (2001–2024, France)	27 cases; median age 4 y	PMID: 41479136
Adult age range	22–77 y; 8M:5F (slight male predominance)	PMID: 29396061

> "We enrolled 540 patients with culture-proven bacterial meningitis during a study period of 30 years (1986-2015), of whom 13 (2.4%) had anaerobic infections." — PMID: 29396061
>
> "24/594 (4%) anaerobic cultures from 14 patients were positive for anaerobes. Only 3 of those patients were diagnosed clinically with anaerobic meningitis, each with predisposing factors" — PMID: 32717475

Population demographics. No ethnic or geographic predilection is established; distribution reflects the prevalence of predisposing conditions (ENT infection, neurosurgery, immunocompromise). Sex ratio is roughly balanced with slight male predominance in adults.

Diagnostics (Section 10)

Core principle: routine aerobic culture misses fastidious anaerobes, and anaerobic culture is low-yield and contamination-prone (many *C. acnes* isolates are contaminants) (PMID: 32717475).

> "In the majority of situations, recovery of C. acnes from CSF or ventricular fluid was regarded as contamination." — PMID: 32717475

Laboratory / CSF. Neutrophilic pleocytosis, elevated protein, low glucose; **CSF lactate** is the most useful discriminator — pooled sensitivity 0.93 (95% CI 0.89–0.96) and specificity 0.96 (95% CI 0.93–0.98) at ~35 mg/dL, generated by bacterial anaerobic metabolism (PMID: 21382412).

> "The pooled test characteristics of CSF lactate were sensitivity 0.93 (95% CI: 0.89-0.96), specificity 0.96 (95% CI: 0.93-0.98)" — PMID: 21382412

Molecular diagnostics (increasingly first-line for anaerobes). 16S rRNA PCR and metagenomic next-generation sequencing (mNGS) recover pathogens when culture fails (PMID: 35692070, PMID: 37775739, PMID: 41479136).

Imaging. Contrast CT/MRI to detect meningeal enhancement, brain abscess, mastoiditis/sinusitis, and cavernous sinus / internal jugular vein thrombosis (PMID: 41330476, PMID: 42568628).

Differential diagnosis. Aerobic bacterial meningitis; aseptic/viral meningitis; tuberculous meningitis; drug-induced aseptic meningitis (e.g., trimethoprim-sulfamethoxazole, PMID: 39540422); primary amebic meningoencephalitis (*Naegleria fowleri*), which mimics bacterial meningitis (PMID: 42426619). Genetic testing (WGS/WES/panels/karyotype/CMA) is **not applicable**.

Outcome / Prognosis (Section 11)

Prognosis is poor, driven by delayed diagnosis and host comorbidity.

Outcome	Value	Source
Adult mortality	30.8% (4/13)	PMID: 29396061
Full recovery among survivors	22.2% (2/9)	PMID: 29396061
Neurological deficits among survivors	77.8% (7/9)	PMID: 29396061
mNGS CNS-anaerobic cohort mortality	3/8	PMID: 35692070
Pediatric *F. necrophorum* deaths	4/27; 20/27 complicated	PMID: 41479136

> "Despite treatment, 30.8% (4/13) of the patients died. Of the nine survivors, 22.2% (2/9) had a full recovery, while the other 77.8% (7/9) had varying degrees of neurological deficits." — PMID: 29396061

Complications: brain abscess, subdural empyema, ventriculitis, cavernous sinus/IJV thrombosis, mastoiditis, Lemierre syndrome with septic pulmonary emboli, hydrocephalus, seizures, sensorineural deficits. **Prognostic factors:** promptness of therapy, source control, host comorbidity, host inflammasome/complement genotype (CARD8, NLRP1, MASP-2), and CSF cytokine levels (PMID: 23053059, PMID: 28086930).

Treatment (Section 12)

Pharmacotherapy — antimicrobial mainstay. Metronidazole (CHEBI:6909; NCIT: Metronidazole) is bactericidal against obligate anaerobes with excellent CSF penetration; it eradicated anaerobic pathogens in CNS infection with 5/6 patients recovering in an early series (PMID: 434970). Modern regimens combine metronidazole with a **third-generation cephalosporin** (ceftriaxone/cefotaxime) for Gram-negative/streptococcal coverage (PMID: 41330476). Alternatives: carbapenems (meropenem) and β-lactam/β-lactamase-inhibitor combinations (piperacillin-tazobactam). Gram-positive anaerobes (*Cutibacterium*, *Parvimonas*) are usually penicillin-susceptible (PMID: 29396061).

> "In all cases the anaerobic pathogens were eradicated. Five of the six patients recovered." — PMID: 434970

Source control / surgery. Surgical excision or drainage of abscess and replacement of infected devices, with prolonged antibiotics (usually 4–8 weeks) when abscess is present (PMID: 28089421, PMID: 32717475).

> "surgical excision or drainage combined with prolonged antibiotics (usually 4-8weeks) remains the treatment of choice." — PMID: 28089421

Adjunctive corticosteroids. Dexamethasone reduces mortality and neurological sequelae in acute bacterial (especially pneumococcal) meningitis when given before/with antibiotics (PMID: 41675892); pediatric data support dexamethasone 0.15 mg/kg q6h × 4 days within 12 h (adjusted 30-day death 6% vs 12%) (PMID: 40113367).

> "Corticosteroids significantly reduce mortality and neurological sequelae in acute bacterial meningitis (particularly pneumococcal), especially when administered before or with antibiotics." — PMID: 41675892

Anticoagulation is used for associated cavernous sinus/IJV thrombosis (Lemierre) though evidence remains debated (PMID: 41330476, PMID: 42568628).

NCIT clinical-intervention terms: Metronidazole, Ceftriaxone, Cefotaxime, Meropenem, Piperacillin-Tazobactam, Dexamethasone, Surgical Drainage.

Prevention (Section 13)

Prevention is **non-vaccine** (no licensed vaccine targets these anaerobes):

- **Primary:** prompt treatment/source control of predisposing ENT, dental, sinus, and abdominal infections; strict neurosurgical and device asepsis; judicious device management.
- **Secondary:** early recognition of complicated ENT infection and Lemierre-associated thrombophlebitis to prevent CNS extension.
- **Tertiary:** prolonged antibiotics, surgical drainage, and device replacement to prevent relapse/complications.
- **Public health:** dental/oral hygiene, timely otitis media/sinusitis management.

No genetic counseling or carrier/newborn screening applies (non-heritable).

Other Species / Natural Disease (Section 14)

Anaerobes cause naturally occurring disease in animals — **necrobacillosis**. *Fusobacterium necrophorum* (NCBI:txid859) is an opportunistic pathogen causing calf diphtheria, foot rot, and liver abscesses in cattle; its **leukotoxin (Lkt)** is the major virulence factor and a protective antigen, with subsp. *necrophorum* more leukotoxic than subsp. *funduliforme* (PMID: 30771459).

> "In animals, it is an opportunistic pathogen frequently associated with necrotic infections, generally called necrobacillosis, such as calf diphtheria, foot rot and liver abscesses in cattle." — PMID: 30771459
>
> "Among several virulence factors, leukotoxin (Lkt) is considered to be a major factor and a protective antigen." — PMID: 30771459

Other implicated anaerobes span species: *B. fragilis* (txid817), *Prevotella*, *Parvimonas micra*, *Clostridium perfringens* (txid1502), *Cutibacterium acnes* (txid1747). Comparative pathology: abscess-forming, tissue-necrotizing anaerobic infection is broadly conserved across mammals, but frank anaerobic *meningitis* is documented predominantly in humans; veterinary anaerobic disease more often manifests as abscesses/soft-tissue necrosis. Zoonotic transmission of these commensal anaerobes is not a recognized route of human meningitis.

Model Organisms (Section 15)

There is no dedicated model of anaerobic meningitis, but **rodent bacterial-meningitis models** (Wistar rats; intracisternal *S. pneumoniae* or *K. pneumoniae*) recapitulate the shared downstream pathophysiology: a TNF-α/IL-1β/IL-6/CINC-1 cytokine surge within 6–24 h, BBB disruption, and long-term cognitive/memory impairment (PMID: 24105840, PMID: 23085269). **Anaerobe-specific animal models** exist for *B. fragilis* abscess pathogenesis (mouse/rat intra-abdominal abscess models establishing capsular polysaccharide as the virulence factor; PMID: 6954635, PMID: 3865350, PMID: 3711690) and for *F. necrophorum* necrobacillosis/leukotoxin (PMID: 30771459). **Limitations:** these models capture the inflammatory cascade and virulence-factor biology but not the specific meningeal invasion route of anaerobes in humans. **Resources:** MGI, RGD (rat), and standard laboratory rodent strains; no organoid/iPSC anaerobic-meningitis model is established.

Mechanistic Model / Interpretation

The unifying model is a **two-hit process**: (Hit 1) a barrier-breaching predisposing lesion allows endogenous anaerobic flora to reach the CSF; (Hit 2) subarachnoid replication triggers a stereotyped, TLR-driven innate inflammatory cascade whose intensity — modulated by pathogen virulence factors and host inflammasome/complement genotype — determines the degree of BBB breakdown and neuronal injury. Anaerobe-specific features graft onto this shared backbone: the *B. fragilis* capsule biases toward walled-off abscess (with weak systemic endotoxicity), while *F. necrophorum* biases toward septic thrombophlebitis (Lemierre) and leukotoxin-mediated tissue necrosis. Therapeutically, the model rationalizes the triad of (1) anaerobe-active antibiotics (metronidazole ± β-lactam) to remove the Hit-2 stimulus, (2) source control to remove the Hit-1 nidus, and (3) corticosteroids to blunt the injurious inflammatory amplification.

Axis	Aerobic/pyogenic meningitis (e.g., pneumococcal)	Anaerobic meningitis
Frequency	Common	Rare (~2.4% of adult culture-proven)
Typical host	Often previously healthy	Predisposed (ENT/neurosurgery/ immunocompromise)
Presentation	Acute, classic triad	Often subacute/atypical
Diagnosis	Routine culture usually positive	Culture low-yield; needs 16S/mNGS
Key virulence	Pneumolysin, capsule	B. fragilis capsule; F. necrophorum leukotoxin
First-line drug	Ceftriaxone (± vancomycin)	Metronidazole + β-lactam

Evidence Base

PMID	Role in report	Key contribution
29396061	Core	Rarity (2.4%), species spectrum, risk factors, mortality/ morbidity
32717475	Core	Low culture yield, contamination, device/procedure risk
41479136	Core	Pediatric *F. necrophorum*, ENT precursor, complications
42568628	Supporting	Defines Lemierre syndrome / thrombophlebitis route
21382412	Diagnostics	CSF lactate diagnostic performance
434970	Treatment	Metronidazole eradication of anaerobic CNS infection
22618789	Mechanism	TLR→NF-κB/MAPK→oxidative injury cascade
23053059	Mechanism/ modifier	CARD8/NLRP1 inflammasome outcome association
28086930	Mechanism/ modifier	MASP-2 complement worsens outcome
24105840	Model organism	Rat cytokine surge, memory impairment
6954635	Virulence	*B. fragilis* capsule = abscess factor
3865350	Virulence	Capsule antiphagocytic
549184	Virulence	Weak *Bacteroides* LPS endotoxicity
3711690	Virulence	LPS dose ~5× CP for abscess
15509818	Phenotype	Frequency of meningitis triad/features
28089421	Treatment/abscess	Source control, 4–8 wk antibiotics, anaerobes lead abscess
41675892	Treatment	Corticosteroid benefit/timing
40113367	Treatment	Pediatric dexamethasone mortality benefit
30771459	Veterinary	Necrobacillosis, leukotoxin
	Diagnostics	mNGS for CNS anaerobes

PMID	Role in report	Key contribution
35692070 / 37775739		
41330476	Case/complication	*F. nucleatum* meningitis + cavernous sinus thrombosis
31838318	Epidemiology	Neonatal anaerobic meningitis rarity

Evidence source types: the mechanism is largely extrapolated from broader bacterial-meningitis literature (human cohort + rodent models + in vitro virulence studies), while anaerobe-specific clinical data come from small human case series and single-/multi-center cohorts.

Limitations and Knowledge Gaps

1. **No anaerobe-specific mechanistic studies of meningitis** — the inflammatory cascade is inferred from pneumococcal/Klebsiella models, not from anaerobe-induced meningitis directly.
2. **Small sample sizes** — the largest dedicated adult series has only 13 cases; pediatric data 27 cases. No prospective trials of anaerobic meningitis specifically exist.
3. **Diagnostic under-ascertainment** — routine practice omits anaerobic culture, so true incidence is likely underestimated; molecular detection is reshaping the epidemiology.
4. **Corticosteroid evidence is indirect** — benefit is established for pneumococcal/general bacterial meningitis, not specifically anaerobic; TBM data show genotype-dependent heterogeneity (PMID: 41540106).
5. **No genetic architecture** — beyond a few inflammasome/complement modifiers, host susceptibility genetics are unexplored.
6. **Optimal antibiotic duration, and the role/timing of anticoagulation** in Lemierre-associated CNS disease remain undefined.

Proposed Follow-up Experiments / Actions

1. **Multicenter registry with routine mNGS/16S** on all culture-negative meningitis CSF to establish true incidence and species distribution of anaerobic meningitis.
2. **Prospective validation of CSF lactate + molecular panels** as a rapid rule-in pathway for anaerobic meningitis, with cost-effectiveness analysis.
3. **Develop an anaerobe-specific meningitis model** (e.g., intracisternal *F. necrophorum* or *B. fragilis*) to directly test the TLR/inflammasome cascade and the effect of capsule/leukotoxin knockouts on CNS injury.
4. **Genotype-stratified adjunctive corticosteroid evaluation** (CARD8/NLRP1/MASP-2) in bacterial meningitis cohorts including anaerobic cases, mirroring the LTA4H-stratified TBM approach.
5. **Structured outcome study** of anticoagulation timing in anaerobic meningitis complicated by cavernous sinus/IJV thrombosis.
6. **Preventive audit** of ENT/odontogenic source control and neurosurgical device asepsis protocols to quantify modifiable risk.

Report compiled from 5 investigation iterations, 11 confirmed findings, and 56 reviewed papers. Evidence is graded by source type (human clinical cohort/case series, model organism, in vitro virulence, computational/meta-analytic). Anaerobic meningitis is a non-heritable, acquired infectious disease; sections pertaining to Mendelian genetics, inheritance, and genetic screening are marked not applicable throughout.
